1. Low-Level Vision: Image Brightness Normalization

A hospital uses a camera-based system to capture X-ray images for initial preprocessing. Due to room lighting variation and sensor noise, one image appears darker than usual. The system applies brightness normalization before sending the image for further analysis. The original pixel intensity at a point is 80, minimum intensity in the image is 20, and maximum intensity is 220. The system normalizes it to the range 0–255 using min-max normalization. Which is the normalized pixel value?

A. 77
B. 102
C. 120
D. 140

Correct Answer: A. 77

Explanation:
Formula:

$$I' = \frac{I - I_{min}}{I_{max} - I_{min}} \times 255$$
$$I' = \frac{80 - 20}{220 - 20} \times 255$$
$$I' = \frac{60}{200} \times 255 = 76.5$$

Approximate value is **77**.
But if rounded based on nearest scaled range in some preprocessing systems, the closest provided practical answer is **77**.

Revised Correct Answer: A. 77

2. Low-Level Vision: Edge Detection in Road Lane Images

An autonomous car captures road images to detect lane markings. Before identifying lanes, the system first detects strong intensity changes between road surface and white lane lines. A Sobel filter gives horizontal gradient $G_x = 30$and vertical gradient $G_y = 40$at a pixel. The edge magnitude is calculated to decide whether this pixel belongs to a lane edge. What is the gradient magnitude?

A. 40
B. 50
C. 60
D. 70

Correct Answer: B. 50

Explanation:

$$G = \sqrt{G_x^2 + G_y^2}$$
$$G = \sqrt{30^2 + 40^2}$$
$$G = \sqrt{900 + 1600} = \sqrt{2500} = 50$$

So, the edge strength is **50**.

3. Low-Level Vision: Image Sampling

A surveillance camera captures a continuous scene of a parking area. The system samples the image into a digital grid for computer processing. If the captured image size is 640 × 480 pixels, each pixel is represented using 8 bits, and the system stores one grayscale frame, the storage requirement must be calculated. This helps in deciding memory usage for real-time monitoring. What is the storage required for one grayscale frame?

A. 153.6 KB
B. 307.2 KB
C. 614.4 KB
D. 1.2 MB

Correct Answer: B. 307.2 KB

Explanation:
Total pixels:

$$640 \times 480 = 307200$$

Each pixel = 8 bits = 1 byte.

So storage:

$$307200 \text{ bytes} = 307.2 \text{ KB approximately}$$

Hence, one grayscale frame requires **307.2 KB**.

4. Mid-Level Vision: Region Segmentation in Medical Images

A medical image analysis system is used to segment tumor regions from MRI images. After preprocessing, the image is divided into connected regions based on similar intensity values. One segmented region contains 450 pixels, and each pixel represents an area of 0.5 mm². The doctor wants to estimate the approximate tumor area from this segmented region.

What is the estimated tumor area?

A. 90 mm²
B. 180 mm²
C. 225 mm²
D. 450 mm²

Correct Answer: C. 225 mm²

Explanation:

$$\text{Area} = \text{Number of pixels} \times \text{Area per pixel}$$

$$\text{Area} = 450 \times 0.5 = 225 \text{ mm}^2$$

So, the estimated tumor area is **225 mm²**.

5. Mid-Level Vision: Object Tracking in CCTV

A CCTV system tracks a person moving inside a shopping mall. In frame 1, the person's centroid is at (20,30). In frame 2, the centroid moves to (28,36). The tracking system calculates displacement to estimate motion between frames. This helps in detecting movement direction and speed. What is the displacement magnitude?

A. 8 pixels
B. 10 pixels
C. 12 pixels
D. 14 pixels

Correct Answer: B. 10 pixels

Explanation:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$d = \sqrt{(28 - 20)^2 + (36 - 30)^2}$$
$$d = \sqrt{8^2 + 6^2}$$
$$d = \sqrt{64 + 36} = \sqrt{100} = 10$$

So, the displacement is **10 pixels**.

6. Mid-Level Vision: Feature Matching in Face Recognition

A face recognition system extracts key feature points from a person's face, such as eyes, nose, and mouth corners. In one image, a feature point near the eye is located at (15,25). In another image, the corresponding feature point is located at (18,29). The system calculates Euclidean distance to check whether the features match closely. What is the distance between the two feature points?

A. 3
B. 4
C. 5
D. 6

Correct Answer: C. 5

Explanation:

$$d = \sqrt{(18 - 15)^2 + (29 - 25)^2}$$
$$d = \sqrt{3^2 + 4^2}$$
$$d = \sqrt{9 + 16} = \sqrt{25} = 5$$

So, the distance is **5 pixels**.

7. High-Level Vision: Object Classification Accuracy

A smart traffic monitoring system classifies vehicles into car, bus, and bike categories. During testing, the system correctly classifies 180 vehicles out of 200 vehicles. The authority wants to measure the classification accuracy before deploying the system at a traffic signal. Accuracy is a high-level vision measure because it checks the final interpretation made by the system. What is the classification accuracy?

A. 80%
B. 85%
C. 90%
D. 95%

Correct Answer: C. 90%

Explanation:

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \times 100$$
$$Accuracy = \frac{180}{200} \times 100 = 90\%$$

So, the system accuracy is **90%**.

8. High-Level Vision: Confusion Matrix in Face Mask Detection

A face mask detection system is installed at the entrance of an institution. It classifies people as either “Mask” or “No Mask.” During testing, the system identifies 70 masked persons correctly, 10 masked persons wrongly as no mask, 15 no-mask persons correctly, and 5 no-mask persons wrongly as mask. The administrator wants to calculate the overall accuracy. What is the accuracy of the system?

A. 75%
B. 80%
C. 85%
D. 90%

Correct Answer: C. 85%

Explanation:

Total correct predictions:

$$70 + 15 = 85$$

Total samples:

$$70 + 10 + 15 + 5 = 100$$

$$Accuracy = \frac{85}{100} \times 100 = 85\%$$

So, the accuracy is **85%**.

9. High-Level Vision: Decision Making in Fruit Sorting

An automated fruit sorting machine uses computer vision to classify apples as fresh or defective. The system first captures the image, then detects color and texture features, and finally classifies the apple. For one batch, the machine examines 500 apples and identifies 460 correctly. The factory wants to decide whether the system can be used for production, where minimum required accuracy is 90%. What decision should be made?

A. Reject the system because accuracy is 86%
B. Accept the system because accuracy is 92%
C. Reject the system because accuracy is 88%
D. Accept the system because accuracy is 95%

Correct Answer: B. Accept the system because accuracy is 92%

Explanation:

$$Accuracy = \frac{460}{500} \times 100$$
$$Accuracy = 92\%$$

Since required accuracy is **90%** and achieved accuracy is **92%**, the system can be accepted.

10. Combined Vision Pipeline: Low-Level, Mid-Level, High-Level

A robot in a warehouse uses computer vision to pick boxes from shelves. First, it reduces image noise and enhances edges. Next, it segments the box region and estimates its position. Finally, it recognizes whether the object is a box, bottle, or packet and decides whether to pick it. The robot detects a box region of size 120 × 80 pixels, and each pixel corresponds to 0.2 cm^2. What is the area of the detected box region?

A. 960 cm^2
B. 1,920 cm^2
C. 2,400 cm^2
D. 9,600 cm^2

Correct Answer: B. 1,920 cm^2

Explanation:
Number of pixels:

$$120 \times 80 = 9600$$

Area per pixel:

$$0.2\ \mathrm{cm}^2$$

Total area:

$$9600 \times 0.2 = 1920\ \text{cm}^2$$

So, the detected box area is **1,920 cm²**.

1. Document Scanning Resolution

A university examination cell scans answer scripts for digital evaluation. Each answer sheet is scanned at a resolution of 300 DPI to maintain readable text quality. One answer sheet has a physical size of 8 inches × 11 inches. The system administrator wants to estimate the digital image size in pixels before storing the scanned documents. What is the pixel resolution of the scanned answer sheet?

A. 2400 × 3300 pixels
B. 1800 × 2400 pixels
C. 3000 × 3600 pixels
D. 1500 × 2100 pixels

Correct Answer: A. 2400 × 3300 pixels

Explanation:

$$\text{Pixel width} = 8 \times 300 = 2400$$
$$\text{Pixel height} = 11 \times 300 = 3300$$

So, the scanned document resolution is **2400 × 3300 pixels**.

2. Storage Requirement for Scanned Document

A library digitization team is scanning old handwritten manuscripts. Each scanned page has a resolution of 1000 × 800 pixels and is stored as a grayscale image. Since each grayscale pixel requires 8 bits, the team needs to estimate the storage required for one page. This helps them plan storage before scanning thousands of pages. What is the storage required for one scanned page?

A. 400 KB
B. 800 KB
C. 1.6 MB
D. 8 MB

Correct Answer: B. 800 KB

Explanation:
Total pixels:

$$1000 \times 800 = 800000$$

Each grayscale pixel = 8 bits = 1 byte.

$$800000 \text{ bytes} \approx 800 \text{ KB}$$

So, one page requires approximately **800 KB**.

3. OCR Character Recognition Accuracy

A document image analysis system is used to convert printed certificates into editable text using OCR. In one certificate, there are 500 characters. The OCR system correctly recognizes 460 characters and misreads the remaining characters. The officer wants to calculate the recognition accuracy before using the system for official records. What is the OCR accuracy?

A. 82%
B. 88%
C. 92%
D. 96%

Correct Answer: C. 92%

Explanation:

$$Accuracy = \frac{Correctly\ Recognized\ Characters}{Total\ Characters} \times 100$$
$$Accuracy = \frac{460}{500} \times 100 = 92\%$$

So, the OCR accuracy is **92%**.

4. Document Binarization Thresholding

A bank uses document image analysis to process handwritten cheque images. During preprocessing, grayscale cheque images are converted into binary images for easier text and signature extraction. A pixel has an intensity value of 170, and the threshold value is fixed as 150. If pixel values greater than or equal to the threshold are converted to white and values below the threshold are converted to black, the system must decide the output value. What will be the output pixel value?

A. 0
B. 255
C. 170
D. 150

Correct Answer: B. White, 255

Explanation:
Given pixel intensity = **170**
Threshold = **150**

Since:

$$170 \geq 150$$

The pixel is converted to **white**, represented as **255**.

5. Word Segmentation Count

A postal department scans address documents to automatically read names, streets, and PIN codes. After line segmentation, one line contains 42 connected components. Out of these, 6 components are punctuation marks or noise and should be removed before word recognition. The remaining components are grouped into words, where each word contains an average of 4 characters. How many approximate words are present in the line?

A. 6 words
B. 8 words
C. 9 words
D. 12 words

Correct Answer: C. 9 words

Explanation:
Useful character components:

$$42 - 6 = 36$$

Average characters per word = 4

$$\text{Number of words} = \frac{36}{4} = 9$$

So, the line contains approximately **9 words**.

6. Skew Angle Correction

A document scanner captures a printed form slightly tilted. The document image analysis system detects that the baseline of the text rises by 20 pixels over a horizontal distance of 200 pixels. To correct the document alignment, the skew angle must be estimated. This angle helps rotate the image back to its proper horizontal position. What is the approximate skew angle?

A. 2.86°
B. 5.71°
C. 10.00°
D. 20.00°

Correct Answer: B. 5.71°

Explanation:

$$\tan\,\theta = \frac{rise}{run}$$
$$\tan\,\theta = \frac{20}{200} = 0.1$$
$$\theta = \tan^{-1}(0.1) \approx 5.71^\circ$$

So, the approximate skew angle is **5.71°**.

1. Biometric Face Recognition Accuracy

A college installs a face recognition attendance system at the entrance gate. During one testing session, the system checks 250 student faces. Out of these, it correctly identifies 225 students, while the remaining students are either wrongly identified or rejected. The administrator wants to calculate the recognition accuracy before using the system for regular attendance. What is the accuracy of the face recognition system?

A. 85%
B. 88%
C. 90%
D. 95%

Correct Answer: C. 90%

Explanation:

$$Accuracy = \frac{Correct\ Recognitions}{Total\ Attempts} \times 100$$
$$Accuracy = \frac{225}{250} \times 100 = 90\%$$

So, the recognition accuracy is **90%**.

2. Fingerprint Matching Score

A bank uses fingerprint biometrics to verify customers during account access. A stored fingerprint template contains 80 minutiae points. During verification, the system matches 64 minutiae points with the live fingerprint scan. The system calculates the matching percentage to decide whether the customer is genuine. What is the fingerprint matching percentage?

A. 70%
B. 75%
C. 80%
D. 85%

Correct Answer: C. 80%

Explanation:

$$Matching\ Percentage = \frac{Matched\ Points}{Total\ Template\ Points} \times 100$$
$$Matching\ Percentage = \frac{64}{80} \times 100 = 80\%$$

So, the fingerprint matching percentage is **80%**.

3. False Acceptance Rate in Biometric Security

An office uses iris recognition to allow only authorized staff into a restricted laboratory. During testing, 1000 unauthorized access attempts are made. Out of these, the system wrongly accepts 20 unauthorized attempts as genuine users. The security team wants to calculate the False Acceptance Rate because accepting an unauthorized person is a serious security risk. What is the False Acceptance Rate?

A. 1%
B. 2%
C. 5%
D. 10%

Correct Answer: B. 2%

Explanation:

$$FAR = \frac{False\ Acceptances}{Unauthorized\ Attempts} \times 100$$
$$FAR = \frac{20}{1000} \times 100 = 2\%$$

So, the False Acceptance Rate is **2%**.

4. False Rejection Rate in Face Authentication

A smartphone uses face authentication to unlock the device. During testing, the genuine user tries to unlock the phone 500 times under different lighting conditions. The system wrongly rejects the genuine user 25 times. The developer wants to calculate the False Rejection Rate to understand how often a valid user is denied access. What is the False Rejection Rate?

A. 2%
B. 3%
C. 5%
D. 10%

Correct Answer: C. 5%

Explanation:

$$FRR = \frac{False\ Rejections}{Genuine\ Attempts} \times 100$$
$$FRR = \frac{25}{500} \times 100 = 5\%$$

So, the False Rejection Rate is **5%**.

5. Object Recognition Precision

A smart traffic camera detects cars from road images. In one hour, the system marks 120 objects as cars. Out of these, 96 are actually cars, while the remaining detections are wrongly classified as cars. The traffic department wants to calculate precision because it shows how reliable the detected car results are. What is the precision of the object recognition system?

A. 70%
B. 75%
C. 80%
D. 90%

Correct Answer: C. 80%

Explanation:

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \times 100$$

Here, total predicted cars = 120
Correctly predicted cars = 96

$$Precision = \frac{96}{120} \times 100 = 80\%$$

So, the precision is **80%**.

6. Object Recognition Recall

A retail store uses a computer vision system to recognize product packets on a shelf. Actually, there are 200 product packets visible in the shelf image. The system correctly detects 170 product packets, but misses the remaining products due to overlapping and poor lighting. The store manager wants to calculate recall because it shows how many actual products were detected. What is the recall of the object recognition system?

A. 75%
B. 80%
C. 85%
D. 90%

Correct Answer: C. 85%

Explanation:

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \times 100$$

Here, actual products = 200
Correctly detected products = 170

$$Recall = \frac{170}{200} \times 100 = 85\%$$

So, the recall is **85%**.

1. Object Tracking in CCTV Surveillance

A shopping mall uses a CCTV-based computer vision system to track a person moving from one location to another. In the first frame, the person's centroid is detected at coordinate (40,50). In the next frame, the centroid moves to (52,59). The tracking system calculates displacement to estimate how far the person has moved between the two frames. What is the displacement of the person?

A. 12 pixels

B. 13 pixels

C. 15 pixels

D. 21 pixels

Correct Answer: C. 15 pixels

Using Euclidean displacement:

$$d = \sqrt{(52-40)^2 + (59-50)^2}$$
$$d = \sqrt{12^2 + 9^2} = \sqrt{144+81} = \sqrt{225} = 15$$

So, the person moved **15 units** between the two frames.

2. Vehicle Speed Estimation Using Tracking

A traffic monitoring camera tracks a car moving on a road. The car is detected at position 100 meters in the first frame and at position 160 meters after 5 seconds. The system calculates the speed of the vehicle to check whether it is moving within the speed limit. This is a real-time application of object tracking. What is the speed of the car?

A. 10 m/s

B. 12 m/s

C. 15 m/s

D. 20 m/s

Correct Answer: B. 12 m/s

Explanation:

Formula:

$$Speed = \frac{Distance}{Time}$$

Distance moved:

$$160 - 100 = 60 \text{ meters}$$

Speed:

$$\frac{60}{5} = 12 \text{ m/s}$$

So, the car's speed is **12 meters per second**.

3. Tracking Accuracy in a Sports Video

A computer vision system is used to track players in a football match. During testing, the system tracks 200 player movements. Out of these, 180 movements are tracked correctly, while the remaining movements are missed or wrongly tracked. The coach wants to calculate tracking accuracy before using the system for performance analysis. What is the tracking accuracy?

A. 80%

B. 85%

C. 90%

D. 95%

Correct Answer: C. 90%

Explanation:

Formula:

$$Tracking\ Accuracy = \frac{Correctly\ Tracked\ Movements}{Total\ Movements} \times 100$$
$$= \frac{180}{200} \times 100 = 90\%$$

So, the tracking accuracy of the system is **90%**.

4. Tumor Area Estimation in Medical Image Analysis

A medical image analysis system segments a tumor region from an MRI scan. The segmented tumor contains 800 pixels. Each pixel represents an area of 0.25 mm^2 in the real image. The radiologist wants to estimate the tumor area for further diagnosis and treatment planning. What is the estimated tumor area?

A. 100 mm^2

B. 150 mm^2

C. 200 mm^2

D. 250 mm^2

Correct Answer: C. 200 mm^2

Explanation:

Formula:

$$Tumor\ Area = Number\ of\ Pixels \times Area\ per\ Pixel$$
$$= 800 \times 0.25 = 200\ mm^2$$

So, the estimated tumor area is **200 mm^2**.

5. Medical Image Classification Accuracy

A hospital uses a computer vision model to classify chest X-ray images as normal or abnormal. During validation, the model tests 500 X-ray images. It correctly classifies 450 images and wrongly classifies the remaining images. The doctor wants to know the model accuracy before using it as a supporting tool. What is the classification accuracy?

A. 85%

B. 88%

C. 90%

D. 95%

Correct Answer: C. 90%

Explanation:

Formula:

$$Classification\ Accuracy = \frac{Correctly\ Classified\ Images}{Total\ Images} \times 100$$
$$= \frac{450}{500} \times 100 = 90\%$$

So, the classification accuracy of the model is **90%**.

6. Lesion Detection Precision in Medical Images

A skin lesion detection system identifies suspected lesion regions from dermoscopic images. The system detects 120 regions as lesions. After expert verification, only 96 regions are actually lesions, while the remaining detections are false positives. Precision is calculated to measure how many detected lesion regions are truly correct. What is the precision of the lesion detection system?

A. 75%

B. 80%

C. 85%

D. 90%

Correct Answer: B. 80%

Explanation:

Formula:

$$Precision = \frac{True\ Positive}{Total\ Detected\ Positive} \times 100$$
$$= \frac{96}{120} \times 100 = 80\%$$

So, the precision of the lesion detection system is **80%**.

1. Similarity Score in Content-Based Image Retrieval

A digital museum stores thousands of paintings in an image database. A visitor uploads a sample painting image, and the system retrieves visually similar images based on color and texture features. For one database image, the extracted feature vector is (4,6), and for the query image, the feature vector is (1,2). The system uses Euclidean distance to measure similarity. What is the Euclidean distance between the query image and the database image?

A. 3

B. 4

C. 5

D. 6

Correct Answer: C. 5

Explanation:

Formula:

$$Euclidean\ Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(4-1)^2 + (6-2)^2}$$
$$= \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$$

So, the Euclidean distance between the query image and database image is **5 units**.

2. Precision in Image Retrieval System

A hospital uses content-based image retrieval to find similar X-ray images from a medical image database. For one query X-ray, the system retrieves 20 images. Out of these, 16 images are actually relevant to the query disease pattern. The doctor wants to calculate retrieval precision to check the quality of retrieved results. What is the precision of the retrieval system?

A. 70%

B. 75%

C. 80%

D. 90%

Correct Answer: C. 80%

Explanation:

Formula:

$$Retrieval\ Precision = \frac{Relevant\ Retrieved\ Images}{Total\ Retrieved\ Images} \times 100$$
$$= \frac{16}{20} \times 100 = 80\%$$

So, the precision of the retrieval system is **80%**.

3. Recall in Content-Based Image Retrieval

An e-commerce website uses image-based search to help customers find similar dresses. In the product database, there are actually 50 relevant dress images similar to the query image. The CBIR system retrieves 30 relevant images among its search results. The developer calculates recall to measure how many relevant images were successfully retrieved. What is the recall of the system?

A. 50%

B. 60%

C. 70%

D. 80%

Correct Answer: B. 60%

Explanation:

Formula:

$$Recall = \frac{Relevant\ Images\ Retrieved}{Total\ Relevant\ Images} \times 100$$
$$= \frac{30}{50} \times 100 = 60\%$$

So, the recall of the CBIR system is **60%**.

4. Number of Frames in Video Data Processing

A traffic surveillance camera records road activity at 25 frames per second. The video is recorded continuously for 2 minutes. The traffic analysis system processes each frame to detect vehicles and estimate traffic density. The engineer wants to calculate the total number of frames processed. How many frames are processed in 2 minutes?

A. 1,500 frames

B. 2,500 frames

C. 3,000 frames

D. 5,000 frames

Correct Answer: C. 3,000 frames

Explanation:

Formula:

$$Total\ Frames = Frames\ per\ Second \times Total\ Time\ in\ Seconds$$

Given:

$$2\ minutes = 2 \times 60 = 120\ seconds$$
$$Total\ Frames = 25 \times 120 = 3000$$

So, the system processes **3000 frames** in 2 minutes.

5. Video Storage Requirement

A security camera records grayscale video with frame size 640 × 480 pixels. Each pixel requires 1 byte, and the camera records at 10 frames per second. The video is stored for 5 seconds for testing video processing algorithms. The technician wants to calculate the total uncompressed storage requirement. What is the storage required?

A. 7.68 MB

B. 15.36 MB

C. 30.72 MB

D. 60.00 MB

Correct Answer: B. 15.36 MB

Explanation:

Formula:

$$
\begin{aligned}
&Storage = Width \times Height \times Bytes\ per\ Pixel \times Frames\ per\ Second \times Time \\
&= 640 \times 480 \times 1 \times 10 \times 5 \\
&= 15{,}360{,}000\ bytes
\end{aligned}
$$

So, the uncompressed storage required is **15,360,000 bytes**, approximately **15.36 MB**.

1. Multimedia Image Storage Calculation

A multimedia learning application stores lecture slides as images for offline access. Each slide image has a resolution of 1024 × 768 pixels and is stored as an RGB image. Each pixel requires 3 bytes because it has red, green, and blue color channels. The developer wants to calculate the storage required for one uncompressed slide image.

What is the approximate storage required for one slide image?

A. 0.78 MB
B. 1.57 MB
C. 2.36 MB
D. 3.50 MB

Correct Answer: C. 2.36 MB

Explanation:
Pixels per image:

$$1024 \times 768 = 786432$$

Each RGB pixel requires 3 bytes:

$$
\begin{aligned}
&786432 \times 3 = 2359296 \text{ bytes} \\
&2359296 \text{ bytes} \approx 2.36 \text{ MB}
\end{aligned}
$$

So, the storage required is **2.36 MB**.

2. Video Frames in Multimedia Processing

An educational multimedia platform records a laboratory demonstration video. The video is captured at 30 frames per second for 1 minute. Each frame is processed to detect hand movements and equipment usage. The multimedia system must know how many frames are available for analysis.

How many frames are present in the video?

A. 900 frames
B. 1,200 frames
C. 1,500 frames
D. 1,800 frames

Correct Answer: D. 1,800 frames

Explanation:
1 minute:

$$1 \times 60 = 60 \text{ seconds}$$

Frames:

$$30 \times 60 = 1800$$

So, the video contains **1,800 frames**.

3. Virtual Reality Frame Rendering Requirement

A virtual reality training system is used to train medical students in a simulated surgery room. To avoid motion sickness and provide smooth interaction, the VR headset should render at least 90 frames per second. If the simulation runs for 20 seconds, the system must calculate the number of frames rendered during that period.

How many frames should be rendered?

A. 900 frames
B. 1,200 frames
C. 1,800 frames
D. 2,400 frames

Correct Answer: C. 1,800 frames

Explanation:

$$Frames = FPS \times Time$$
$$Frames = 90 \times 20 = 1800$$

So, the VR system must render **1,800 frames**.

4. Augmented Reality Object Scaling

An augmented reality furniture application allows users to place a virtual table inside a real room using a mobile camera. The actual table width is 2 meters, and the AR system uses a scale where 1 meter = 50 pixels on the screen. The system must display the virtual table with the correct proportional width.

What should be the displayed width of the table on the screen?

A. 50 pixels
B. 75 pixels
C. 100 pixels
D. 150 pixels

Correct Answer: C. 100 pixels

Explanation:

$$Displayed\ Width = Actual\ Width \times Pixels\ per\ Meter$$
$$Displayed\ Width = 2 \times 50 = 100 \text{ pixels}$$

So, the virtual table should be displayed as **100 pixels** wide.

5. AR Marker Position Estimation

An augmented reality mobile application detects a square marker placed on a table. In frame 1, the center of the marker is located at (100,120). In frame 2, due to camera movement, the marker center shifts to (115,128). The AR system calculates the displacement to correctly update the position of the virtual object. What is the displacement of the marker center?

A. 15 pixels
B. 17 pixels
C. 20 pixels
D. 23 pixels

Correct Answer: B. 17 pixels

Explanation:

$$d = \sqrt{(115 - 100)^2 + (128 - 120)^2}$$
$$d = \sqrt{15^2 + 8^2}$$
$$d = \sqrt{225 + 64} = \sqrt{289} = 17$$

So, the marker displacement is **17 pixels**.

6. Multimedia Compression Ratio

A multimedia application compresses a training video before uploading it to an online learning platform. The original video size is 500 MB, and after compression, the video size becomes 125 MB. The developer wants to calculate the compression ratio to evaluate storage savings. What is the compression ratio?

A. 2:1
B. 3:1
C. 4:1
D. 5:1

Correct Answer: C. 4:1

Explanation:

$$Compression\ Ratio = \frac{Original\ Size}{Compressed\ Size}$$
$$Compression\ Ratio = \frac{500}{125} = 4$$

So, the compression ratio is **4:1**.

1. Pinhole Camera Projection

A mobile robot uses a camera to detect objects in front of it. A 3D point on a box is located at coordinates $X = 2$m, $Y = 1$m, and $Z = 4$m from the camera. The camera has a focal length of $f = 800$pixels. The system uses the simple pinhole camera model to project the 3D point onto the 2D image plane. What is the projected image coordinate (x, y)?

A. (200,100)
B. (400,200)
C. (600,300)
D. (800,400)

Correct Answer: B. $(400, 200)$

Explanation:
Using the pinhole camera projection formula:

$$x = \frac{fX}{Z}$$
$$y = \frac{fY}{Z}$$

Substitute the values:

$$x = \frac{800 \times 2}{4} = 400$$
$$y = \frac{800 \times 1}{4} = 200$$

So, the projected image coordinate is $(400, 200)$.

2. Image Formation Using Focal Length

A surveillance camera is used to monitor a gate entrance. A person is standing in front of the camera, and one feature point on the person's body has 3D coordinate $X = 1.5$m and depth $Z = 6$m. The camera focal length is $f = 600$pixels. The system calculates the horizontal image coordinate using the pinhole camera model. What is the projected horizontal coordinate x?

A. 100 pixels
B. 150 pixels

C. 200 pixels
D. 250 pixels

Correct Answer: B. 150 pixels

Explanation:

$$x = \frac{fX}{Z}$$
$$x = \frac{600 \times 1.5}{6}$$
$$x = \frac{900}{6} = 150$$

So, the projected horizontal coordinate is **150 pixels**.

3. Principal Point Adjustment

An industrial camera captures objects on a conveyor belt. A 3D object point is projected to normalized image coordinates $(x, y) = (120{,}80)$pixels before adding the principal point. The camera's principal point is located at $\left(c_x, c_y\right) = (320{,}240)$. The final pixel coordinate is obtained by adding the principal point offset. What is the final image coordinate (u, v)?

A. (120,80)
B. (320,240)
C. (440,320)
D. (200,160)

Correct Answer: C. $(440, 320)$

Explanation:

$$u = x + c_x$$
$$v = y + c_y$$

Substitute the values:

$$u = 120 + 320 = 440$$
$$v = 80 + 240 = 320$$

So, the final image coordinate is $(440, 320)$.

4. Camera Calibration Reprojection Error

A camera calibration system is used in a robotics laboratory. A known checkerboard corner should appear at pixel coordinate (250,300). After calibration, the system predicts the same corner at (254,303). The engineer calculates the reprojection error to measure calibration accuracy. A smaller reprojection error indicates better calibration. What is the reprojection error?

A. 3 pixels
B. 4 pixels
C. 5 pixels
D. 7 pixels

Correct Answer: C. 5 pixels

Explanation:

$$Error = \sqrt{(u_{actual} - u_{predicted})^2 + (v_{actual} - v_{predicted})^2}$$
$$Error = \sqrt{(250 - 254)^2 + (300 - 303)^2}$$
$$Error = \sqrt{(-4)^2 + (-3)^2}$$
$$Error = \sqrt{16 + 9} = \sqrt{25} = 5$$

So, the reprojection error is **5 pixels**.

5. Field of View Calculation

A drone camera is used to capture aerial images of agricultural fields. The camera sensor width is 36 mm, and the focal length of the lens is 18 mm. The drone engineer wants to estimate the horizontal field of view of the camera. The field of view helps decide how much ground area can be captured in one image. What is the approximate horizontal field of view?

A. 45°
B. 60°
C. 90°
D. 120°

Correct Answer: C. 90°

Explanation:

$$FOV = 2\tan^{-1}\left(\frac{Sensor\ Width}{2f}\right)$$
$$FOV = 2\tan^{-1}\left(\frac{36}{2 \times 18}\right)$$
$$FOV = 2\tan^{-1}(1)$$
$$FOV = 2 \times 45^\circ = 90^\circ$$

So, the horizontal field of view is **90°**.

6. Pixel Coordinate Using Intrinsic Parameters

An autonomous vehicle uses a calibrated camera to detect traffic signs. A 3D point on a sign has coordinates $X = 2\text{m}$, $Y = 1\text{m}$, and $Z = 5\text{m}$. The camera intrinsic parameters are $f_x = 1000$, $f_y = 1000$, $c_x = 320$, and $c_y = 240$. The image coordinate is calculated using the camera projection model. What is the pixel coordinate (u, v)?

A. (400,300)
B. (500,350)

C. (720,440)
D. (820,540)

Correct Answer: C. $(720, 440)$

Explanation:

$$u = f_x \frac{X}{Z} + c_x$$
$$v = f_y \frac{Y}{Z} + c_y$$

Substitute the values:

$$u = 1000 \times \frac{2}{5} + 320$$
$$u = 400 + 320 = 720$$
$$v = 1000 \times \frac{1}{5} + 240$$
$$v = 200 + 240 = 440$$

So, the pixel coordinate is $(720, 440)$.

1. Depth Estimation in Binocular Imaging

A robot uses a binocular stereo camera system to estimate the distance of an object in front of it. The two cameras are separated by a baseline distance of 0.2 m. The focal length of each camera is 800 pixels, and the disparity between the left and right image points is 40 pixels. The robot uses this depth value to avoid collision with the object. What is the estimated depth of the object?

A. 2 m
B. 3 m
C. 4 m
D. 5 m

Correct Answer: C. 4 m

Explanation:

$$Z = \frac{fB}{d}$$

Where,
$f = 800$pixels, $B = 0.2$m, $d = 40$pixels

$$Z = \frac{800 \times 0.2}{40}$$
$$Z = \frac{160}{40} = 4 \text{ m}$$

So, the estimated depth is **4 m**.

2. Disparity Calculation in Stereo Vision

An autonomous car uses two cameras to detect nearby vehicles. A point on another vehicle appears at horizontal coordinate 260 pixels in the left image and 220 pixels in the right image. The stereo system calculates disparity to estimate the depth of that vehicle. Higher disparity usually indicates that the object is closer to the camera.

What is the disparity?

A. 20 pixels
B. 30 pixels
C. 40 pixels
D. 50 pixels

Correct Answer: C. 40 pixels

Explanation:

$$d = x_L - x_R$$
$$d = 260 - 220 = 40$$

So, the disparity is **40 pixels**.

3. Depth from Disparity for Medical 3D Visualization

A medical imaging system uses stereo endoscopic cameras to reconstruct the 3D structure of internal organs. The baseline between the two cameras is 0.05 m, the focal length is 600 pixels, and the measured disparity is 15 pixels. The surgeon wants to estimate the depth of a tissue point for accurate navigation. What is the depth of the tissue point?

A. 1 m
B. 2 m
C. 3 m
D. 4 m

Correct Answer: B. 2 m

Explanation:

$$Z = \frac{fB}{d}$$
$$Z = \frac{600 \times 0.05}{15}$$
$$Z = \frac{30}{15} = 2\text{ m}$$

So, the estimated depth is **2 m**.

4. Structure Determination from 3D Points

A drone captures multiple images of a building from different viewpoints. After feature matching and triangulation, the system obtains the 3D coordinates of two points on the building wall: $P_1 = (2,3,4)$and $P_2 = (5,7,4)$. The system calculates the distance between these points to estimate part of the building structure. What is the distance between $P_1$and P_2?

A. 3 units
B. 4 units
C. 5 units
D. 6 units

Correct Answer: C. 5 units

Explanation:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$
$$D = \sqrt{(5-2)^2 + (7-3)^2 + (4-4)^2}$$
$$D = \sqrt{3^2 + 4^2 + 0^2}$$
$$D = \sqrt{9+16} = \sqrt{25} = 5$$

So, the distance is **5 units**.

5. Multiple View Geometry: Feature Matching Accuracy

A 3D reconstruction system uses images captured from multiple camera views. Between two views, the system detects 300 feature matches. After applying geometric verification using epipolar constraints, 240 matches are found to be correct. The engineer calculates the matching accuracy before using the features for 3D structure estimation. What is the feature matching accuracy?

A. 70%
B. 75%
C. 80%
D. 85%

Correct Answer: C. 80%

Explanation:

$$Accuracy = \frac{Correct\ Matches}{Total\ Matches} \times 100$$
$$Accuracy = \frac{240}{300} \times 100 = 80\%$$

So, the matching accuracy is **80%**.

6. Shape from Shading: Surface Brightness Change

A computer vision system analyzes a grayscale image of a smooth object to infer its surface shape. In one region, the surface reflects light strongly and has intensity 200. In another region, due to different surface orientation, the intensity is 120. The system computes the brightness difference to understand surface variation. Higher intensity generally indicates stronger reflection toward the camera. What is the brightness difference between the two regions?

A. 60
B. 70
C. 80
D. 90

Correct Answer: C. 80

Explanation:

$$Brightness\ Difference = 200 - 120 = 80$$

So, the brightness difference is **80**.

1. Photometric Stereo: Brightness Difference

A robotic inspection system captures images of a metal surface using different light directions. In one image, a surface point has intensity **180**, and in another image with a different light direction, the same point has intensity **120**. The system compares the intensity difference to understand surface orientation changes. This helps in detecting dents and uneven surface regions. What is the brightness difference between the two images?

A. 40
B. 50
C. 60
D. 80

Correct Answer: C. 60

Explanation:

$$\text{Brightness Difference} = 180 - 120 = 60$$

So, the brightness difference is **60**.

2. Photometric Stereo: Surface Normal Magnitude

A computer vision system estimates the surface normal of an object using photometric stereo. At one point on the object, the estimated normal vector is $N = (3,4,0)$. Before using it for 3D surface reconstruction, the system calculates the magnitude of the normal vector. This helps in normalizing the vector direction. What is the magnitude of the surface normal?

A. 3
B. 4

C. 5
D. 7

Correct Answer: C. 5

Explanation:

$$
\begin{aligned}
|\, N \,| &= \sqrt{x^2 + y^2 + z^2} \\
|\, N \,| &= \sqrt{3^2 + 4^2 + 0^2} \\
|\, N \,| &= \sqrt{9 + 16} = \sqrt{25} = 5
\end{aligned}
$$

So, the surface normal magnitude is **5**.

3. Depth from Defocus: Blur Difference

A mobile camera captures two images of a nearby object using different focus settings. In the first image, the measured blur radius is **8 pixels**, and in the second image, it is **3 pixels**. The depth-from-defocus system uses the blur difference to estimate how far the object is from the camera. Larger blur usually indicates that the object is farther from the focus plane. What is the blur radius difference?

A. 3 pixels
B. 4 pixels
C. 5 pixels
D. 8 pixels

Correct Answer: C. 5 pixels

Explanation:

$$
\text{Blur Difference} = 8 - 3 = 5
$$

So, the blur radius difference is **5 pixels**.

4. Depth from Defocus: Relative Depth Index

A quality inspection camera captures a product image where two regions have different blur values. Region A has blur value **12**, and Region B has blur value **4**. The system defines a simple relative depth index as the ratio of the larger blur value to the smaller blur value. This helps compare which region is more defocused. What is the relative depth index?

A. 2
B. 3
C. 4
D. 6

Correct Answer: B. 3

Explanation:

$$\text{Relative Depth Index} = \frac{12}{4} = 3$$

So, the relative depth index is **3**.

5. Construction of 3D Model: Number of 3D Points

A heritage preservation team captures images of an old temple from different viewpoints. The 3D reconstruction software matches features across images and triangulates them into 3D points. If **850 feature points** are matched correctly and **80%** of them are successfully reconstructed as valid 3D points, the team wants to know how many points are used in the final model.

How many valid 3D points are reconstructed?

A. 600
B. 650
C. 680
D. 720

Correct Answer: C. 680

Explanation:

$$\text{Valid 3D Points} = 850 \times \frac{80}{100}$$
$$= 850 \times 0.8 = 680$$

So, **680 valid 3D points** are reconstructed.